Forecasting Takaful Application Demand

Student Practical Assignment • Microsoft Excel 365 & Copilot

Assessment	Individual or group practical (as directed by instructor)
Total marks	100 marks
Suggested time	90–120 minutes
Dataset	Prudential_BSN_Granular_AI_Forecasting_Dataset.xlsx (synthetic training data)

Scenario

You are an operations analyst supporting a fictional takaful provider. Management needs an evidence-based forecast of incoming application volumes to plan staffing, processing capacity and service-level performance. You have been given application-level records and aggregated daily, weekly and monthly histories. Your task is to use Microsoft Excel and, where available, Microsoft 365 Copilot to analyse historical patterns, forecast demand and communicate recommendations.

Important: All data are fictional. Do not treat the dataset as actual Prudential BSN information. Forecasts are estimates, not guaranteed outcomes.

Dataset and rules

- Training period: January 2023–September 2025. Forecast period: October–December 2025.
- Use Applications_Raw, Daily_History, Weekly_History and Monthly_History for analysis.
- Do not inspect Trainer_Holdout or Actual_After_Reveal until the instructor authorizes the evaluation stage.
- Use Excel formulas, PivotTables and charts to verify numerical claims made by Copilot. If Copilot is unavailable, complete the analysis using native Excel features.
- Save a separate working copy. Document assumptions and any data transformations.

Part A — Explore and prepare the data (20 marks)

A1. Inspect the raw application records. State the number of records, the date range, the number of branches and the number of product categories. Explain the difference between application-level and aggregated time-series data. (6 marks)

A2. Create a monthly application-volume PivotTable from Applications_Raw. Check that the monthly totals agree with Monthly_History. Describe any differences and how you resolved them. (8 marks)

A3. Identify at least two data-quality or forecasting-readiness checks, including treatment of missing dates, partial periods or irregular time intervals. (6 marks)

Part B — Analyse trends and seasonality (20 marks)

B1. Produce a labelled line chart showing monthly application volume for the complete training period. Describe the overall direction of the trend and identify the highest and lowest months. (8 marks)

B2. Analyse whether daily volumes differ between weekdays and weekends. Present one appropriate table or chart. (6 marks)

B3. Discuss possible seasonal or campaign-related effects. Distinguish observed patterns from unproven causal explanations. (6 marks)

Part C — Generate and justify forecasts (25 marks)

C1. Use Data → Forecast Sheet or FORECAST.ETS to predict the total number of applications in October, November and December 2025. Provide your three forecast values in a table. (12 marks)

C2. Explain your chosen forecasting method, input data frequency, seasonality setting (if any) and at least two assumptions or limitations. (8 marks)

C3. Present a chart that clearly distinguishes historical actuals from forecast values. Include a forecast confidence interval if available. (5 marks)

Part D — Evaluate accuracy (15 marks)

Only after completing Part C, obtain permission to reveal actual application volumes for October–December 2025.

D1. Enter actuals into Forecast_Challenge and calculate absolute error, mean absolute error (MAE) and mean absolute percentage error (MAPE). Show the formulas or explain their calculation. (9 marks)

D2. Identify the month with the largest forecasting error. Suggest two plausible reasons and one improvement to your forecasting workflow. (6 marks)

Part E — Copilot and management decision-making (20 marks)

E1. Write at least two prompts to Copilot: one for analysing historical patterns and one for evaluating your forecast. Include the prompts and summarize the AI responses. (8 marks)

E2. Verify at least two numerical or interpretive claims made by Copilot against Excel results. Correct any inaccurate or unsupported statements. If Copilot is unavailable, describe the prompts you would use and demonstrate the verification in Excel. (6 marks)

E3. Prepare a short management recommendation (150–200 words) explaining how the forecast could inform staffing and processing capacity, together with uncertainties that management should consider. (6 marks)

Required submission

- Completed Excel workbook containing your PivotTables, formulas, forecast values, evaluation metrics and charts.
- Brief report (2–4 pages, PDF or Word) with labelled figures, answers to Parts A–E and your management recommendation.
- Copilot prompts and a brief record of how you verified its output.
- Clear identification of all assumptions and any manual data corrections.

Assessment rubric

Criterion	Marks
A. Data exploration and preparation	20
B. Trend and seasonality analysis	20
C. Forecast generation and justification	25
D. Forecast evaluation	15
E. Copilot validation and management recommendations	20
TOTAL	100

Academic and responsible AI use

You may use Copilot as an analytical assistant, but you are responsible for checking its calculations, chart interpretations and recommendations. Do not invent results, present assumptions as facts or submit AI-generated conclusions without verification. Use only the supplied synthetic dataset and institutionally approved tools.